

# vanilla-extract

## Usage, Methods and Validation Results

Version 0.2.0 | verification run 2026-09-16 | MIT licensed | [github.com/brandonsbutler-ai/vanilla-extract](https://github.com/brandonsbutler-ai/vanilla-extract)

This document exists so you do not have to take the claims on faith. It states what the tool does, how it is checked, what the checks measured, and -- the part most such documents omit -- what the numbers do **not** prove.

Every figure below is produced by a script in this repository. You can run them yourself; the commands are at the end.

## 1. What it is

vanilla-extract extracts text and named fields from documents, and produces a reviewable table from a folder of them. It reads PDF, DOCX, PPTX, XLSX, ODT, RTF, EML, MBOX, HTML, XML, CSV, TSV, JSON and plain text, plus ZIP archives containing any of those.

It has **no third-party dependencies**. Not "few" -- none. That is not a stylistic preference: it means the tool installs in an environment where adding a dependency requires an approval process, and it means there is no supply chain under it to audit.

## 2. Usage

### Read one document

```
vanilla contract.pdf
```

### A folder into a spreadsheet

```
vanilla --batch invoices/ --csv results.csv --exceptions skipped.csv
```

Two tables, always. `results.csv` has a row per document read. `skipped.csv` names every document that could **not** be read and why. The second file is the important one, and section 5 explains why.

### Pull named fields

```
vanilla --batch invoices/ --csv results.csv \
  --field "invoice_no=Invoice\s*#\s*([A-Z0-9-]+)" \
  --field "total=Total\s*:\s*\$?([0-9,]+\.[0-9]{2})"
```

### Or let it find the fields itself

```
vanilla --batch invoices/ --recognize --report review.html --csv results.csv
```

`--recognize` reads label/value pairs out of the documents and ranks them by how many documents carry each label, so the columns come from your corpus rather than from a guess. Each column is typed -- money, date, email, phone, percent, identifier -- by a conservative detector.

Measured on a four-document sample, with no pattern written by hand:

|                 |               |            |                              |
|-----------------|---------------|------------|------------------------------|
| claim number    | 4 docs (100%) | identifier | e.g. CL-4401                 |
| settlement date | 4 docs (100%) | date_iso   | e.g. 2026-08-02              |
| insurer         | 4 docs (100%) | text       | e.g. Ridgeway Mutual         |
| terms           | 4 docs (100%) | text       | e.g. Net 30                  |
| amount          | 4 docs (100%) | money      | e.g. \$2,145.00              |
| contact         | 4 docs (100%) | email      | e.g. claims@ridgeway.example |

## Review and correct before you trust it

--report review.html writes a single HTML file -- no server, no external assets, nothing uploaded. Every value is editable in place, each document's source text is one click away so a suspect value can be checked against the page it came from, the exceptions table is shown alongside rather than buried, and a button exports the corrected table as CSV. Because nothing leaves the machine, a review of your own confidential documents stays on your machine.

## The datasheet: what each file was when it arrived

```
vanilla --batch invoices/ --datasheet state.html --csv out.csv
```

A searchable, sortable page recording each file's state as found: size, modified, accessed and created times, rwx permissions and octal mode, owner and group, symlink status and target, hard-link count, inode, filesystem type, and the extraction outcome. A file that could not be read still appears, with its state and the reason.

Three caveats are built into the output rather than left to the reader:

- **Creation time is not universally available.** Windows records it, macOS and the BSDs expose it, and on Linux it exists only on some filesystems. Every row states where its value came from. The inode-change time is reported under its own name and never presented as creation.
- **\*\*NTFS, exFAT and network mounts report ownership and permissions from mount options, not from the files.\*\*** Those rows are marked not reliable, and the page says why.
- **A file inside a ZIP carries its own timestamp and mode**, which can predate the archive by years. Those come from the archive's central directory.

## Keep the original and every correction

```
vanilla --batch invoices/ --recognize --workspace case01/ --report review.html --csv out.csv
vanilla --workspace case01/ --import-csv corrected.csv
vanilla --workspace case01/ --verify
```

```
case01/
  manifest.json      run metadata and a SHA-256 for every artefact
  originals/         your documents as received
  extracted/         what the tool read, per document, never edited
  revisions/         each corrected table, appended, never replaced
```

Filing a correction states exactly what moved:

```
revision 2: 4 rows, 2 cell(s) changed from the previous revision
  INV-1001.txt: customer: 'Northwind Traders' -> 'Northwind Traders LLC'
  INV-1002.txt: total: '$433.00' -> '$433.50'
```

This answers a question that otherwise has no answer three months later: was this value extracted from the document, or typed by a person?

## 3. How it is verified

Three layers, because they catch different things.

| Layer                                | What it is                            | What it catches                                                       |
|--------------------------------------|---------------------------------------|-----------------------------------------------------------------------|
| Unit tests (100)                     | python3 -m unittest discover -s tests | Logic errors in one function, and every past bug as a regression test |
| Benchmark                            | python3 benchmark.py <dir>            | Extraction *quality*, scored against an independent implementation    |
| End-to-end verification (156 checks) | python3 verify_e2e.py                 | Whether the product does what its documentation says                  |

The third layer is the unusual one. Unit tests check units; they cannot tell you the README is wrong. verify\_e2e.py generates a fresh corpus in every supported format, drives the real command-line tool as a user would, and asserts each documented claim against actual output. It prints PASS or FAIL per claim with the evidence and exits non-zero on any

failure.

It is built to be able to say the product does not work. That is the only reason it is worth running -- and it has done so twice, which section 7 covers.

## 4. Extraction quality, measured

PDF output is scored against [poppler](https://poppler.freedesktop.org/)'s `pdftotext`, a mature C++ implementation with full font handling, by **token recall**: the fraction of the reference's words that vanilla\_extract also produced, counted as a multiset so repetition is not rewarded.

poppler is the reference, not the target. It is allowed to win; the point is knowing by how much.

**Corpus A -- 210 tool-generated PDFs** (fpdf2, LibreOffice Writer):

```
token recall : mean 1.000  median 1.000  min 1.000
>= 0.99      : 210/210 (100.0%)
```

**Corpus B -- 86 commercial PDFs** (Adobe PDF Library, Mac Quartz; illustrated, multi-column, 100-578 pages):

```
files measured : 53
token recall   : mean 0.873  median 0.912  min 0.007  max 0.961
>= 0.95        : 4/53   (7.5%)
>= 0.90        : 35/53  (66.0%)
>= 0.50        : 51/53  (96.2%)

encrypted, refused with a clear error: 33 (reported separately, not scored)
```

Read plainly: **excellent on PDFs produced by software, good but not perfect on complex commercial ones, and never quite matching poppler on the hardest.** If you need the last few percent on illustrated multi-column layouts, use poppler. If you need no dependencies, or you need the field recognition and the audit trail, use this.

## Throughput

Over 14 unencrypted PDFs of 5 MB and up:

|                                           | pages/second |
|-------------------------------------------|--------------|
| Median                                    | 101          |
| Fastest                                   | 179          |
| Slowest measured (image-heavy, four runs) | 42           |

So: **40 pages per second or better, typically around 100, on an otherwise idle machine.** Throughput is wall-clock and measures the machine as much as the code -- the same files drop to 28-35 pages/s at a load average of 4 -- so the verifier reports these numbers alongside the load average that produced them rather than asserting a floor that a busy server would fail for the wrong reason.

A file is read fully into memory, so peak usage tracks document size. One PDF content stream is inflated with a 64 MB ceiling.

## 5. It fails loudly

The worst behaviour a text extractor can have is returning something that looks like a result. An empty string reads as an empty document. Mojibake poisons a dataset silently. Both get a specific, named error instead:

| Condition                   | Behaviour                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encrypted PDF               | EncryptedPDF. Commercial PDFs are routinely encrypted with an <i>*empty*</i> user password purely to set permission flags -- readable in a viewer, still encrypted on disk. 33 of the 86 files in corpus B are like this. |
| Fonts with no character map | UndecodableText, raised only when <i>*nothing*</i> survives. A page mixing an unmapped decorative heading with readable body text keeps the body.                                                                         |
| Unrecognised format         | UnsupportedFormat                                                                                                                                                                                                         |
| Decompression bomb          | ArchiveTooLarge, raised before allocating                                                                                                                                                                                 |

Read fine, contains no text      Reported in the exceptions table as `no_text_found` -- almost always a scan with no text layer. Not a blank row.

In batch mode none of these abort the run. Each becomes a row in the exceptions table, so a folder of 400 documents with 3 bad ones yields 397 results and 3 named failures.

## 6. Hostile input

The tool's entire job is reading files supplied by someone else, so the input is untrusted by definition. These were tested, not assumed:

| Attack                           | Result                                                                                                                                                                                                                                                |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ZIP decompression bomb           | Refused. Members are vetted against declared size and compression ratio before a byte is read, with a running budget for the archive. A 199 KB file declaring 200 MB is refused in 0.000 s.                                                           |
| PDF decompression bomb           | Refused. Inflation is bounded; a 597 KB file declaring 600 MB peaks near 128 MB instead of 616 MB.                                                                                                                                                    |
| XXE (external entity)            | Not exploitable. Verified against <code>file:///etc/passwd</code> .                                                                                                                                                                                   |
| Billion laughs                   | Bounded by the XML parser a few levels in. Amplification, not denial of service.                                                                                                                                                                      |
| Catastrophic backtracking        | None found. Recognizer patterns and the RTF tokenizer stay linear on 5,000-character labels, 20,000 spaces, 50,000 control words.                                                                                                                     |
| Spreadsheet formula injection    | Neutralized. A document containing <code>=cmd ' /C calc!A0</code> would otherwise execute when the client opened the CSV. Cells leading with <code>= + - @</code> tab or CR are prefixed, in both the file writer and the report's in-browser export. |
| Script injection into the report | Fixed. <code>&lt;/script&gt;</code> inside a document's text closed the report's data block. Now escaped.                                                                                                                                             |
| Path traversal                   | A source label containing <code>../</code> or an absolute path cannot escape a workspace.                                                                                                                                                             |

Limits are generous on purpose. A real 300-page report is large and legitimate; the point is to refuse the absurd, not the merely big -- a 20,000-paragraph document at a 29:1 compression ratio still extracts in 0.02 s.

## 7. What the verification found

A validation report that reports nothing is not evidence of quality; it is evidence of a weak process. Here is what these checks actually caught.

### An independent code review found 15 defects. Two were critical:

- RTF: `{\*\generator}` -- a group Word, WordPad and RichEdit write into *\*every\** file -- caused the rest of the document to be discarded. Essentially all real-world RTF returned an empty string. The existing unit test used a simpler construct and passed.
- HTML: `<meta>` and `<link>` have no closing tag but were treated as containers, which latched a counter open and dropped every byte after them. Every real web page, and every HTML-only email body, returned empty. The existing unit test used HTML with no `<meta>` tag.

Also found and fixed: script injection into the report; spreadsheet formula injection; an unbounded decompression path in the primary format; quadratic time and memory on malformed PDFs; one malformed character-map entry rendering an entire readable PDF undecodable; a field pattern that aborted a whole batch; a re-scan that overwrote its own audit trail; an archive that vanished without appearing in either output table; and a stray line in the test file that made running it directly execute 29 of 100 tests and exit zero.

### The end-to-end verifier then caught two more, including one in the documentation:

- A stated throughput of "50-70 pages per second" was wrong in both directions: the median is 101, but one image-heavy file runs at 42. The claim is now 40 or better, typically 100, and the check asserts the *\*floor\**, because the floor is what you rely on.
- One of its own checks was too strict and failed on correct behaviour. It was corrected to test for a parsed tag rather than for escaped text.

Every one of those is now a named regression test.

## 8. What these results do not prove

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- **\*\*The datasheet records what the filesystem reported, which is not always the truth about the file.\*\*** Timestamps can be set by any program with write access, and a foreign mount synthesizes ownership. The output flags what it cannot vouch for; it does not make the filesystem trustworthy.
- **SHA-256 in a workspace proves integrity, not authorship.** It detects drift. It is not a signature, and a workspace is not tamper-proof against someone who can write to it.
- **Benchmark corpora are not your documents.** Corpus A is software-generated, corpus B is commercial publishing. Neither is a folder of your invoices. Send a sample.
- **No OCR.** A scanned page holds an image, not text, and yields nothing -- reported, not silently blank.
- **Layout is approximated, not reconstructed.** Multi-column pages interleave. This is the main source of the gap against poppler on corpus B.
- **Fonts are keyed by resource name across a document** , not per page. A file reusing one font name for two different fonts on different pages can decode one of them wrong.
- **Encryption is detected, never bypassed.** This tool will not help you read a document you do not have the password for.
- **\*\*156 passing checks means the documented claims hold today, on this machine, for these inputs.\*\*** It does not mean the tool is free of defects. The review above found 15 after the unit tests were green.

## 9. Reproduce all of it

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```
git clone https://github.com/brandonsbutler-ai/vanilla-extract
cd vanilla_extract

python3 -m unittest discover -s tests -v      # 100 unit tests
python3 verify_e2e.py                        # 156 end-to-end claim checks
python3 benchmark.py /path/to/your/pdfs      # quality against pdftotext
```

benchmark.py needs poppler ( pdftotext ) for the comparison. Nothing else needs anything.

Every number in this document comes from those three commands.